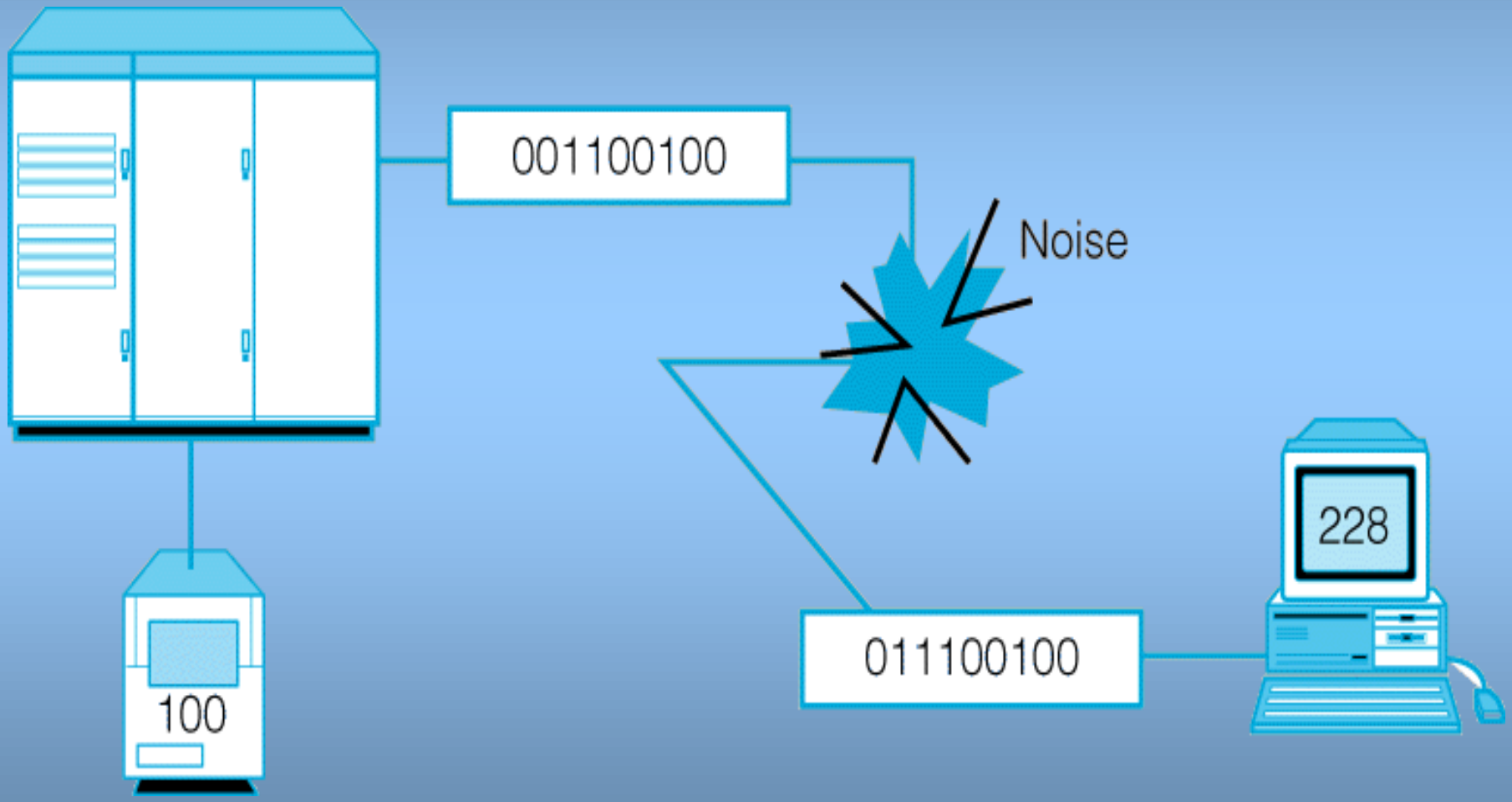
Transmission Errors 1

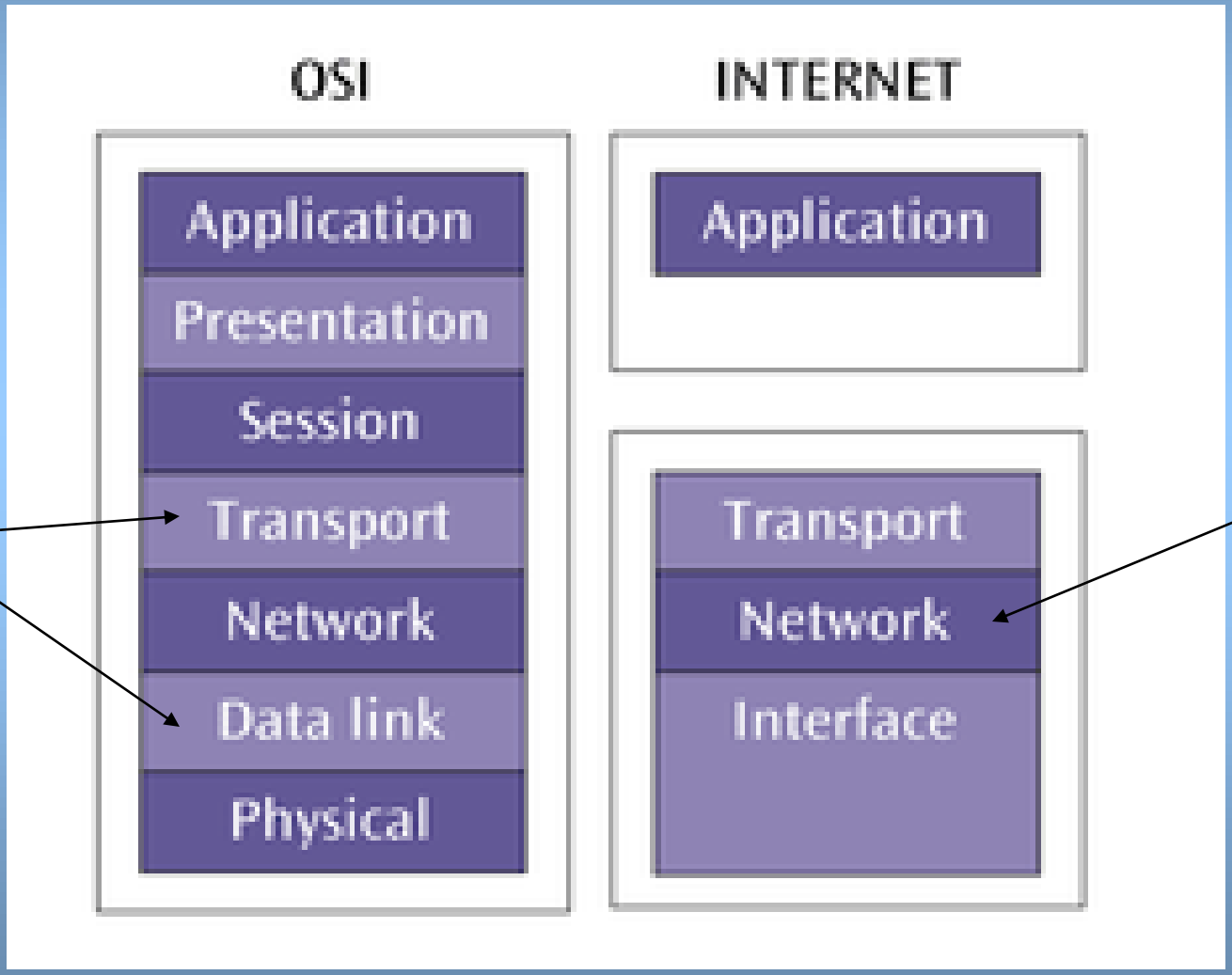
Sunimal Rathnayake
Department of CSE
University of Moratuwa

Reference: Data Communications and Computer Networks: A Business User's Approach by Curt M. White

Transmission Error



OSI vs Internet Model



Error detection in different layers

- Some error detection in other layers
  - Transport layer in OSI
  - Network layer in Internet model
- Focus on the data link layer
  - Independent of the medium or physical layer

Data Link Layer

- Responsible for taking the data and transforming it into a *frame* with header, control and address information.
- Physical path communication
- Error detection
- Error correction
- Resolve competing requests

Overview

- Errors Happen
- Noise and Errors
- Error Prevention
- Error Detection Techniques
- Error Control- Next week

Introduction

Noise is always present.

If a communications line experiences too much noise, the signal will be lost or corrupted.

Communication systems should check for transmission errors.

Once an error is detected, a system may perform some action.

Some systems perform no error control, but simply let the data in error be discarded.

Noise and Errors – White Noise

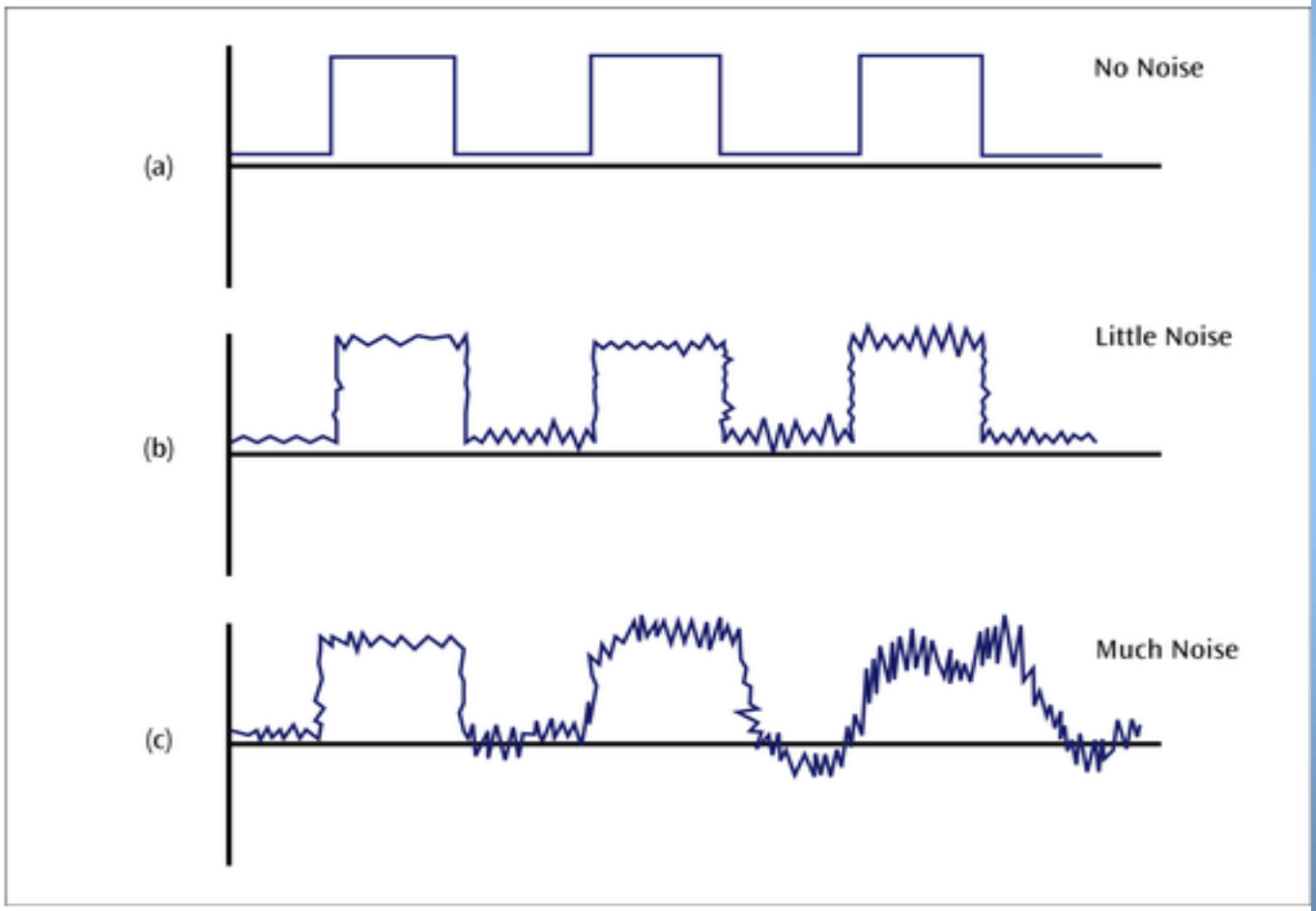
Also known as thermal or Gaussian noise

Relatively constant and can be reduced.

If white noise gets to strong, it can completely disrupt the signal.

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Figure 6-1
White noise as it interferes with a digital signal



Noise and Errors – Impulse Noise

One of the most disruptive forms of noise.

Random spikes of power that can destroy one or more bits of information.

Difficult to remove from an analog signal because it may be hard to distinguish from the original signal.

Impulse noise can damage more bits if the bits are closer together (transmitted at a faster rate).

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Figure 6-2
The affect of impulse noise on a digital signal

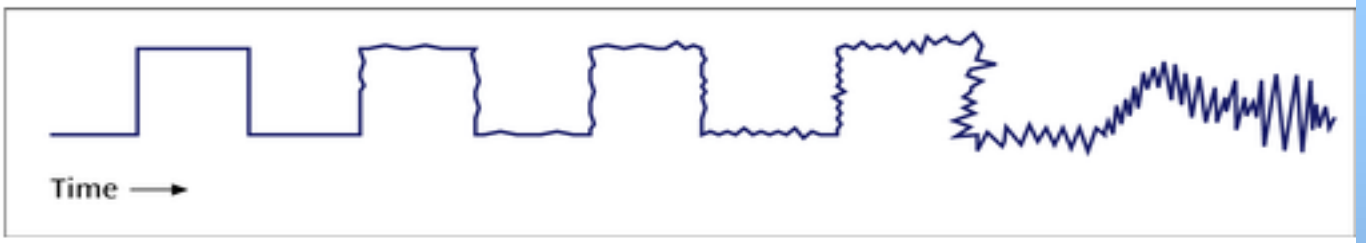
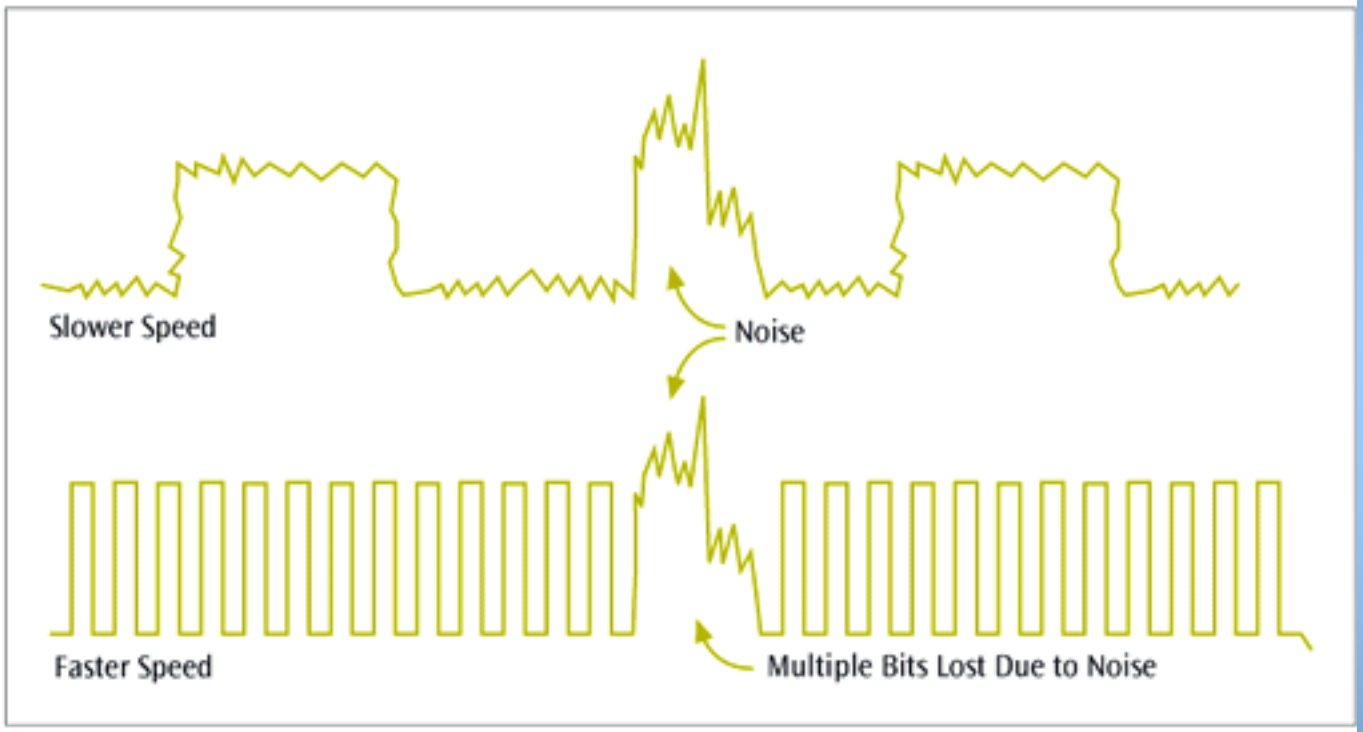


Figure 6-3
Transmission speed and its relationship to noise



Noise and Errors - Crosstalk

Unwanted coupling between two different signal paths.

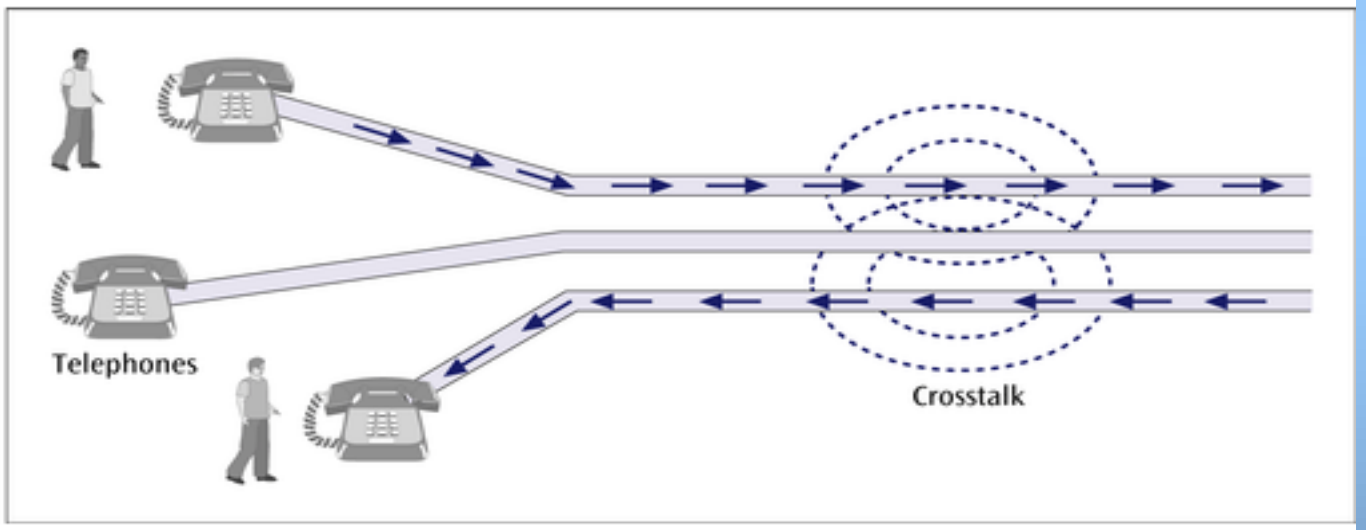
For example, hearing another conversation while talking on the telephone.

Relatively constant and can be reduced with proper measures.

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Figure 6-4
Two telephone circuits experiencing crosstalk



Noise and Errors - Echo

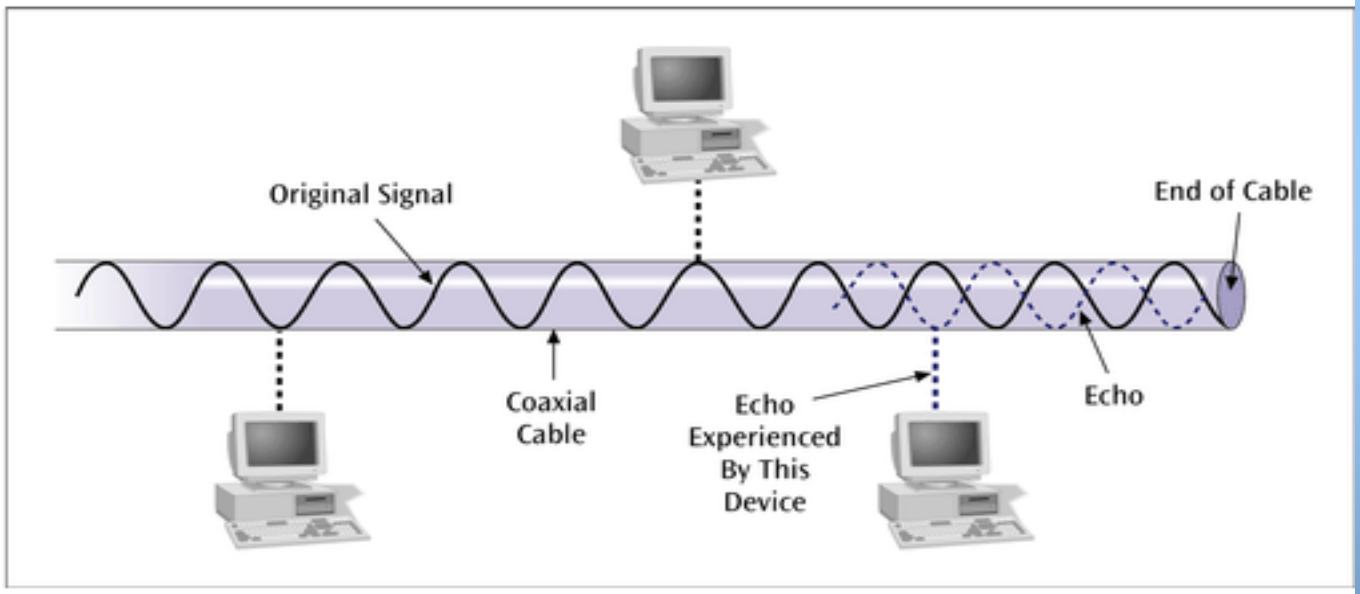
The reflective feedback of a transmitted signal as the signal moves through a medium.

Most often occurs on coaxial cable.

If echo bad enough, it could interfere with original signal.

Relatively constant, and can be significantly reduced.

Figure 6-5
A signal bouncing back at the end of a cable and causing echo



Noise and Errors - Jitter

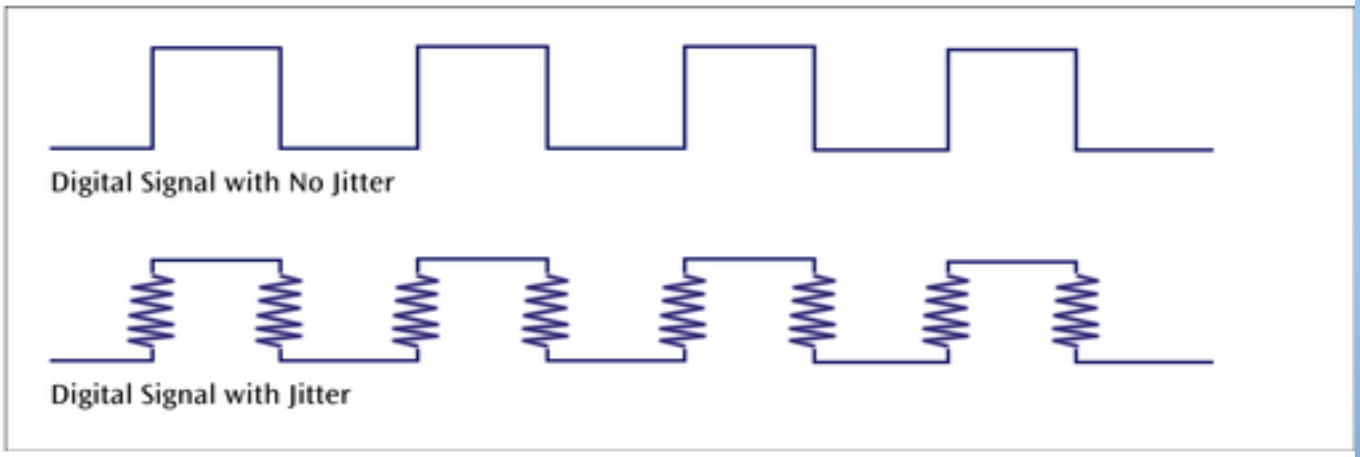
The result of small timing irregularities during the transmission of digital signals.

Occurs when a digital signal is repeated over and over.

If serious enough, jitter forces systems to slow down their transmission.

Reduce jitter - shielding

Figure 6-6
Original digital signal and digital signal with jitter



Noise and Errors – Delay Distortion

Occurs because the velocity of propagation of a signal through a medium varies with the frequency of the signal.
Can be reduced using equalizers

Attenuation

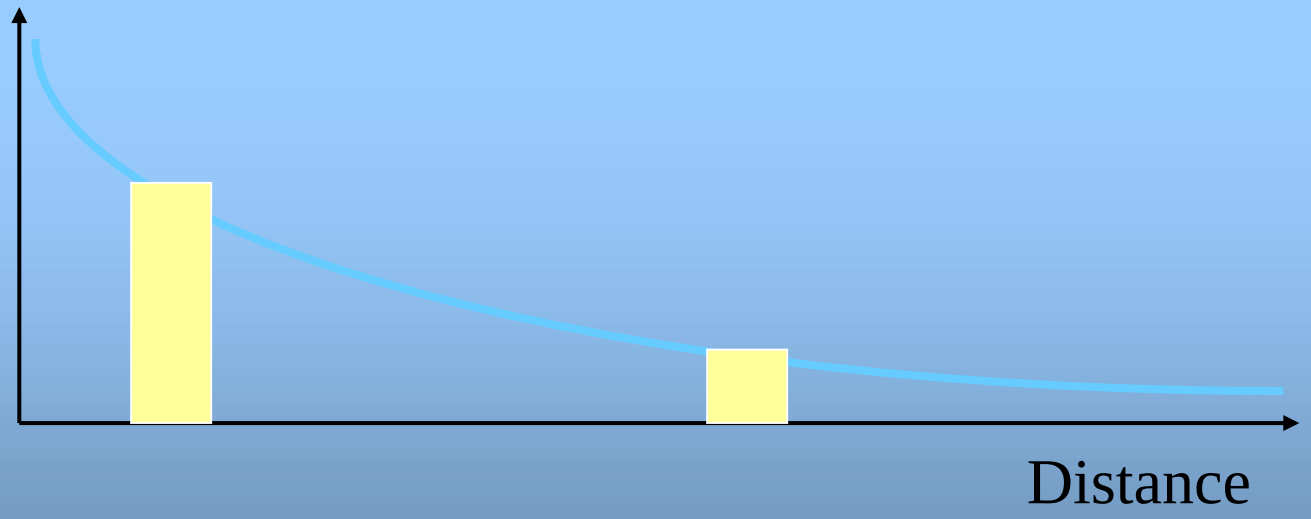
The continuous loss of a signal's strength as it travels through a medium.

Use less lossy medium

Use amplifiers

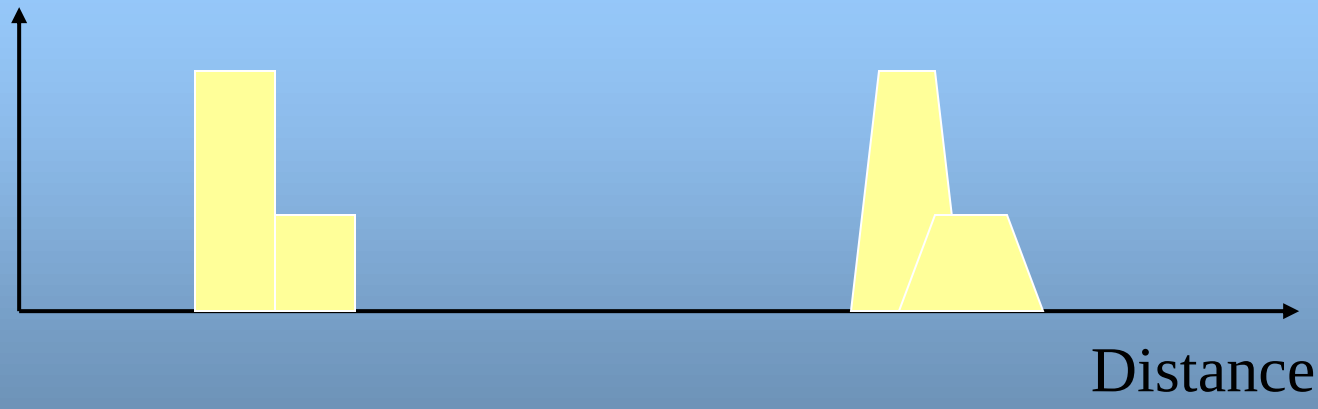
Attenuation

- When signal travels, it gets weaker.
  - If too weak, cannot tell 1s and 0s.



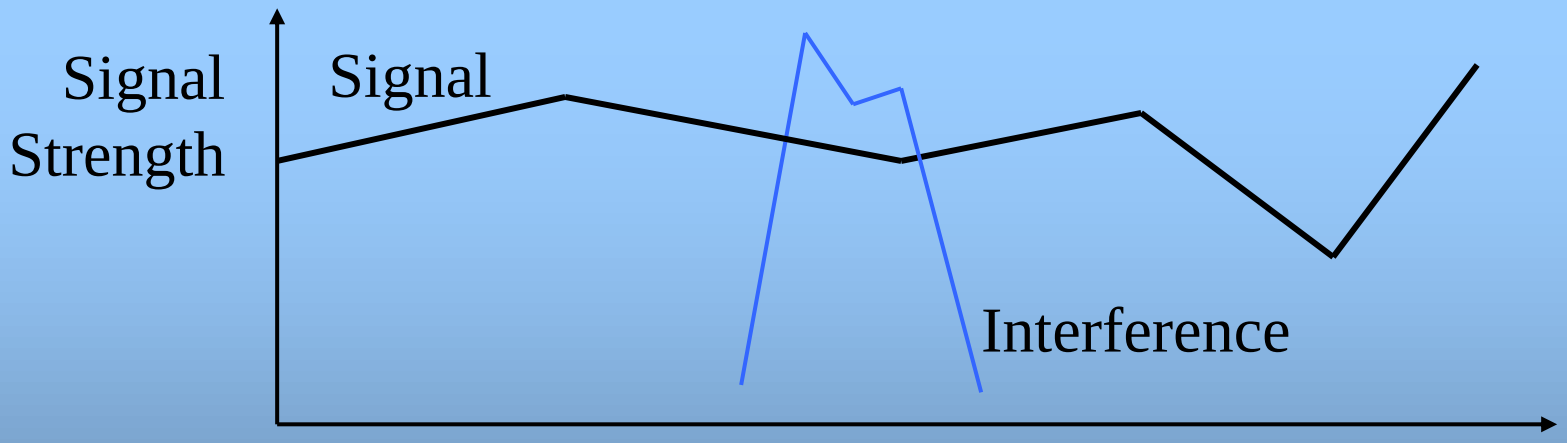
Distortion

- As signal travels, it become distorted.
  - Changes shape
  - Successive bits may merge, making reception difficult



Interference

- Unwanted signal from outside sources
  - Often intermittent, difficult to diagnose



Error Prevention

To prevent errors from happening, several techniques may be applied:

- Proper shielding of cables to reduce interference

- Telephone line conditioning or equalization

- Replacing older media and equipment with new, possibly digital components

- Proper use of digital repeaters and analog amplifiers

- Observe the stated capacities of the media